10TH CLASS COMPUTER SCIENCE — NEW SYLLABUS (PECTAA)

Chapter 5 — Data Science

Solved Exercise | New Book (PECTAA)

Exercise (Solved)

A. Multiple Choice Questions

1. What is the first step in the Data Science Life Cycle (DSLC)?

(a) Collecting Data (b) Visualizing Data
(c) Understanding the Problem (d) Analyzing the Data

Answer: (c) — The Data Science Life Cycle starts with understanding the problem. Before collecting any data, it must be clear what question has to be answered and what the objectives are; only then the right data is collected.

2. Which method is best for collecting opinions from many students?

(a) IoT Devices (b) Survey
(c) Web Scraping (d) Database

Answer: (b) — A survey, for example a form made in Google Forms, is the best method for collecting the opinions of many people, because the same questions are asked from everyone and the answers are collected at one place. IoT devices collect sensor data and web scraping takes data from websites, not opinions.

3. What is "bias" in data?

(a) Very large data size (b) Unfair or one-sided data
(c) Well-cleaned data (d) High-speed data

Answer: (b) — Bias means that the data is one-sided and shows only a part of the truth. For example, if only the boys are surveyed about sports, the opinions of the girls are missed and the result is tilted towards the boys.

4. Which tool is ideal for beginners to create charts?

(a) Tableau (b) Python
(c) Google Sheets (d) Hadoop

Answer: (c) — Google Sheets is a free and simple spreadsheet tool which is ideal for beginners; the data is entered in the cells and a chart is inserted with a few clicks. Tableau and Python are powerful but advanced, and Hadoop is used for big data storage.

5. A relational database stores information in the form of:

(a) Slides (b) Tables
(c) Blocks (d) Layers

Answer: (b) — A relational database stores the information in tables, which are made of rows (records) and columns (fields). For example, a student table has the columns roll_number, name and marks.

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6. Which of the following charts is best for comparing groups?

(a) Pie Chart (b) Bar Chart
(c) Line Chart (d) Area Chart

Answer: (b) — A bar chart compares the quantities of different groups, such as the favourite sports of the students, by showing a bar for every group. A pie chart shows parts of a whole and a line chart shows a trend over time.

7. What is the main goal of analyzing data?

(a) Delete old data (b) Decorate data
(c) Find patterns and make decisions (d) Upload to the internet

Answer: (c) — Analyzing data means carefully examining it to summarise it, compare the groups and find the patterns, so that better decisions can be taken. For example, after analyzing the survey data we come to know that most students prefer cricket.

8. Which of the following tools is an advanced data visualization platform?

(a) MS Paint (b) Google Data Studio
(c) Notepad (d) WhatsApp

Answer: (b) — Google Data Studio is an advanced data visualisation platform which turns the data into reports and interactive charts. MS Paint and Notepad are simple drawing and text tools, and WhatsApp is a messaging application.

9. What should be done before storing data?

(a) Add animations (b) Clean and validate it
(c) Encrypt it (d) Translate it

Answer: (b) — The raw data is often messy, so it must be cleaned and validated before storing it. In this step the duplicates are removed, the spelling and format errors are corrected and the missing values are fixed, so that the stored data is trustworthy.

10. Which one is an example of a data source?

(a) YouTube (b) Survey form
(c) Whiteboard (d) Slide show

Answer: (b) — A data source is the place from which the data is collected, such as a survey form, a database, an API, web scraping or an IoT sensor. A survey form collects the opinions and answers of the people, so it is a data source.

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B. Short Questions

1. What is the importance of understanding a problem before collecting data?

Understanding the problem is the first step of the Data Science Life Cycle, and it is very important because it decides what data is needed and what has to be found out. In this step the problem is written clearly and the objectives are made like a checklist. If the problem is not understood properly, the wrong data will be collected, the wrong questions will be asked and all the later work will be useless. For example, if the school wants to start a new sports club, it must first decide that it needs to know which sports the students like most; only then a survey about the favourite sports is useful.

2. Name any two methods of collecting data and give one example of each.

(i) Survey: the same questions are asked from a group of people, usually with a form such as Google Forms, and their answers are collected at one place. **Example:** a form in which the students select their favourite sport.

(ii) IoT devices (sensors): the data is collected automatically by the sensors. **Example:** smart water meters placed near the school fountains to record how much water the students drink.

Two other methods are **web scraping**, which takes public data from the websites, for example the prices of snacks from online stores, and **APIs**, which give the data of a service directly to the program.

3. What do we mean by “cleaning” data?

Cleaning data means removing the mistakes and the unnecessary parts from the raw data so that the results become trustworthy. The raw data which comes from the surveys and the websites is often messy: it may contain duplicate entries, spelling mistakes, missing values and different formats for the same thing, such as “Cricket” and “cricket”, or “9th” and “9”. In the cleaning step these duplicates are removed, the spellings and the formats are made the same, and the wrong or the empty values are corrected or fixed.

4. Why is it important to validate data before using it?

Validation means checking that the data is complete, correct and relevant to the problem. It is important because the whole analysis depends on the data: if the data is wrong or incomplete, the charts and the conclusions will also be wrong, and the decisions taken from them will be harmful. Validated data is free from the missing values and the mistakes and it matches the problem, so the results can be trusted. For example, if we want to know the favourite snacks of the students of 2025, an old survey of 2020 is not valid for this purpose.

5. What is the purpose of visualizing data using charts?

Data visualization means presenting the data in the form of charts and graphs. Its purpose is to make the data easy to understand, because people understand a picture much faster than a long list of numbers. A good chart shows the patterns at a glance, compares the groups easily and helps to share the findings with others. For example, instead of saying that 12 students like cricket, 5 like football and 3 like badminton, a bar chart shows at once that cricket is the most popular sport.

6. How does bias in data affect results?

Bias means that the data is one-sided and shows only a part of the truth. When biased data is used, the results are also tilted and they do not represent all the people, so the wrong decisions are taken. For example, if only the boys of a school are asked about their favourite sport, the opinions of the girls are missed and the result is towards the boys' choices. To avoid bias, a diverse group of people should be surveyed, the questions should be neutral and not leading, and the answers should be reviewed for strange patterns.

7. Write two steps you will follow to clean a messy survey.

(i) Remove the duplicates and the empty entries: if the same student has filled the form twice, the extra entry is deleted, and the rows in which the important answers are missing are completed or removed.

(ii) Correct the spellings and make the formats the same: the wrong spellings are fixed and the same answer is written in one way, for example “Cricket” and “cricket” are both made “Cricket”, and the class is written either as “9” everywhere or as “9th” everywhere.

8. Name any two tools used for creating charts.

(i) Google Sheets: a free spreadsheet tool which is ideal for the beginners; the data is entered in the cells and a bar, pie or line chart is inserted in a few clicks.

(ii) Google Data Studio: an advanced data visualisation platform which makes interactive reports and charts from the data.

Microsoft Excel and Tableau are other well-known tools for creating charts, and charts can also be drawn with the Python libraries.

C. Long Questions

1. Explain the six steps of the Data Science Life Cycle (DSLC) with one line each.

Data Science is not done in one go; it follows a series of steps which is called the Data Science Life Cycle (DSLC). Its six steps are the following.

Step	One-line explanation
1. Understanding the problem	Write the problem clearly and make the objectives like a checklist, so that it is known what has to be found out.

Step	One-line explanation
2. Data collection	Gather the required data from the right sources, such as surveys (Google Forms), APIs, web scraping and IoT sensors.
3. Data cleaning and validation	Remove the duplicates, fix the errors and the missing values and check that the data is complete, correct and relevant.
4. Data analysis	Examine the data to summarise it, compare the groups and find the patterns with the help of counts, averages and frequencies.
5. Interpretation	Explain what the results mean and what decision or answer they give for the original problem.
6. Data visualization	Present the findings in charts such as bar, pie and line charts, so that everyone can understand them easily.

In some large projects an additional step of **storing the data** is also used, in which the cleaned data is kept in a database with the help of a DBMS and SQL.

2. Give a detailed comparison between Bar Chart, Pie Chart, and Line Chart with examples.

Bar Chart	Pie Chart	Line Chart
It uses rectangular bars of different heights to show the quantity of each group.	It uses a circle divided into slices to show the parts of a whole.	It uses points joined by a line to show how a value changes.
It is best for comparing different groups with one another.	It is best for showing the share of each part in the total, in percentages.	It is best for showing a trend over a period of time.
Example: the number of students who like cricket, football and badminton, shown by three bars.	Example: out of 20 students, 60 % like cricket, 25 % football and 15 % badminton, shown by the slices of a circle.	Example: the marks of a student in the monthly tests of January, February and March, shown by a rising or falling line.
Reading the exact value of each group is easy from the height of the bar.	It becomes difficult to read when there are too many slices.	It clearly shows whether the value is increasing, decreasing or staying the same.

The correct chart is always chosen according to the data. For comparing the groups a bar chart is used, for showing the parts of a whole a pie chart is used, and for showing a change over time a line chart is used.

3. Differentiate between DBMS and SQL.

DBMS	SQL
A Database Management System is the software which is used to create, manage and interact with a database.	SQL (Structured Query Language) is the language which is used to work with the data inside a database.
It stores the data in tables and keeps it organised and safe, and it controls who can see or change the data.	It is used to add, update, delete and retrieve the data from those tables with commands.
Examples: MySQL, SQLite and Microsoft Access.	Main commands: SELECT to get the data, INSERT to add the data, UPDATE to change it and DELETE to remove it.
It is a complete software, so it has to be installed.	It is a language, so it is used inside a DBMS; it is not installed separately.

Example: MySQL is a DBMS which holds the student table. To see the data of that table we write an SQL query inside MySQL:

SQL query

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SELECT * FROM student;
```

4. Write a Query in SQL to get all data of a student whose roll_number is 15 from student table.

The required query is:

Query

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SELECT * FROM student WHERE roll_number = 15;
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Explanation of the query:

(i) SELECT * means that all the columns of the table are to be shown.

(ii) FROM student tells that the data is to be taken from the table named student.

(iii) WHERE roll_number = 15 is the condition, so only that row is shown in which the value of the roll_number column is 15.

For example, if the table also contains the columns name and marks, the query will show the roll number, name and marks of only that one student whose roll number is 15, and the data of all the other students will not appear in the result.